

Capstone Project — Time Series Analysis 2026

Data-Driven Stock Analysis using Time Series Models

Consulting & Analytics Club, IIT Guwahati

Capital: Rs. 10,00,000 Platform: StockGro (NSE live data)

Data: 2020-01-01 to 2026-05-12 Train: 1,434 obs Test: 226 obs

Models	ARIMA($p, 1, q$) (AIC grid, walk-forward), Prophet, Ensemble
Allocation	50% Forecast-return + 50% Inverse-volatility
Metrics	MAPE, RMSE, Directional Accuracy vs. Naive baseline
Diagnostics	Jarque-Bera, Ljung-Box (lag 10), STL decomposition
Portfolio	$\mu_P = +1.07\%$ expected return, $\sigma_P = 21.68\%$ (ann.)

Methodology Overview

The pipeline enforces strict sequential dependency — each stage produces artefacts consumed by the next, ensuring full reproducibility from a single script.

Universe → ADF+Clean → ARIMA/Prophet/Ensemble → STL+Vol → Portfolio → StockGro → Live Eval

Toolchain: `yfinance` (data), `statsmodels` (ARIMA, STL, ADF), `prophet` (Prophet), `scipy` (Jarque–Bera), `sklearn` (RMSE). ARIMA orders were selected automatically via bounded AIC grid search, while Prophet used conservative fixed hyperparameters to reduce overfitting. Ensemble weights were determined through inverse-MAPE weighting, and portfolio allocation combined forecast-return and inverse-volatility approaches.

Task 1 — Stock Universe Selection

Five NSE-listed equities from distinct sectors were selected to maximise sector-level diversification across the portfolio.

Table 1: Selected universe with annualised rolling-30d average volatility (2020–2026)

Ticker	Sector	Avg Ann. Vol.	Regime
RELIANCE.NS	Energy	25.12%	MEDIUM
HDFCBANK.NS	Banking	23.02%	MEDIUM
TATASTEEL.NS	Metals	33.62%	HIGH
APOLLOHOSP.NS	Healthcare	28.85%	HIGH
MARUTI.NS	Auto	25.42%	MEDIUM

RELIANCE.NS (Energy) — macro-energy cycles combined with Jio/retail growth phases produce persistent autoregressive behaviour, a strong ARIMA candidate.

HDFCBANK.NS (Banking) — lowest average volatility in the universe; the HDFC–HDFC Bank merger (Jul 2023) creates a significant structural event that tests whether $d = 1$ differencing absorbs level shifts cleanly.

TATASTEEL.NS (Metals) — highest average volatility; commodity cyclicity and non-Gaussian return tails stress-test residual diagnostics. Its AIC-best order $(0, 1, 0)$ (pure random walk) is a meaningful empirical finding in itself.

APOLLOHOSP.NS (Healthcare) — secular growth with a confirmed UP STL trend slope. Its large absolute price level (Rs. 7,000–8,000) inflates RMSE while leaving MAPE unaffected, so both metrics are reported to avoid misinterpretation.

MARUTI.NS (Auto) — festive-season demand cycles are in principle suited to Prophet’s yearly seasonality component; this inclusion directly tests whether Prophet’s seasonal modelling outweighs its trend-overshoot tendency on financial data.

Task 2 — Data Preprocessing

Acquisition and Cleaning

Adjusted-close prices fetched via `yfinance` for the period 2020-01-01 to 2026-05-12 at daily (1d) interval. Index resampled to business-day frequency and gaps filled with forward-fill then backward-fill:

```
close = close.asfreq("B")
close = close.ffill().bfill()
```

Augmented Dickey–Fuller Stationarity Test

H_0 : unit root present (non-stationary). Rejected at $p < 0.05$.

Table 2: ADF test results — price levels vs. log returns

Ticker	Price Level			Log Return		
	Stat	p	Result	Stat	p	Result
RELIANCE.NS	−2.031	0.2730	Non-stat.	−14.747	< .0001	Stationary
HDFCBANK.NS	−1.773	0.3939	Non-stat.	−17.216	< .0001	Stationary
TATASTEEL.NS	−0.497	0.8926	Non-stat.	−12.389	< .0001	Stationary
APOLLOHOSP.NS	−0.996	0.7548	Non-stat.	− 9.676	< .0001	Stationary
MARUTI.NS	−0.897	0.7889	Non-stat.	−10.079	< .0001	Stationary

All five price-level series fail to reject the unit-root null ($p \geq 0.27$), confirming $I(1)$ behaviour. TATASTEEL’s level statistic of -0.497 ($p = 0.893$) is the weakest, consistent with its high-noise commodity nature. All log-return series reject the null at $p < 10^{-4}$, justifying the fixed $d = 1$ across every ARIMA specification. No additional manual differencing is needed as the ARIMA framework internalises it.

Train / Test Split

- **Train:** 2020-01-01 to 2025-06-30 (1,434 observations per ticker)
- **Test:** 2025-07-01 to 2026-05-12 (226 observations per ticker)

Task 3 — Time Series Forecasting

ARIMA — AIC Grid Search + Walk-Forward Validation

The model order $(p, 1, q)$ was selected by bounded AIC grid search over $p, q \in \{0, 1, 2\}$, keeping only fully converged fits:

$$\text{AIC} = 2k - 2 \ln \hat{L} \quad (p^*, q^*) = \arg \min_{p, q} \text{AIC} \quad (1)$$

A rolling walk-forward scheme was applied on the full 226-step test set: at each step the model is refit on all history up to $t - 1$, a single-step-ahead forecast is issued, then the true observation is appended. An explosion guard prevents degenerate outputs:

```
if pred <= 0 or pred > max(history) * 3:
    pred = history[-1] # fall back to persistence
```

Table 3: ARIMA results — 226-obs walk-forward test set (2025-07-01 to 2026-05-12)

Ticker	Order	MAPE	RMSE	DirAcc	Naive MAPE
RELIANCE.NS	(2, 1, 2)	0.88%	17.70	59.7%	0.89%
HDFCBANK.NS	(2, 1, 2)	0.81%	10.23	51.8%	0.81%
TATASTEEL.NS	(0, 1, 0)	1.19%	3.10	55.8%	1.19%
APOLLOHOSP.NS	(2, 1, 2)	0.86%	91.64	53.1%	0.86%
MARUTI.NS	(2, 1, 2)	1.03%	208.57	50.0%	1.02%

MAPE improvements over the naive baseline are intentionally modest because short-horizon equity prices are highly efficient and difficult to forecast consistently.

TATASTEEL’s best order (0, 1, 0) is a pure random walk: AIC found no (p, q) structure worth retaining. Its Ljung-Box $p = 0.934$ confirms this — no detectable autocorrelation remains. RELIANCE achieves the strongest directional accuracy (59.7%) and MARUTI the weakest (50.0%), consistent with the latter’s noisier recent-period price dynamics.

Facebook Prophet

Configured with `weekly_seasonality=True`, `yearly_seasonality=True`, and conservative `changepoint_prior_scale=0.05`. Predictions on the test set were generated by passing exact test-period dates to prevent look-ahead:

```
fc_test = model.predict(te_df[["ds"]]) # te_df contains only test dates
```

Table 4: Prophet test-set performance — 226-obs window

Ticker	MAPE	RMSE
RELIANCE.NS	14.52%	235.09
HDFCBANK.NS	12.60%	147.54
TATASTEEL.NS	26.60%	54.96
APOLLOHOSP.NS	4.17%	370.52
MARUTI.NS	18.24%	3200.88

Prophet’s piecewise-linear trend overestimates long-run momentum and fails to track the short-horizon mean-reverting fluctuations that dominate equity prices at daily resolution. TATASTEEL reaches 26.6% MAPE, roughly 22× worse than ARIMA on the same series.

Inverse-MAPE Weighted Ensemble

$$\hat{y}_t^{\text{ens}} = \frac{(1/M_{\text{AR}}) \hat{y}_t^{\text{AR}} + (1/M_{\text{Pr}}) \hat{y}_t^{\text{Pr}}}{1/M_{\text{AR}} + 1/M_{\text{Pr}}} \quad M_m = \text{test MAPE of model } m \quad (2)$$

Table 5: Ensemble test-set performance

Ticker	Ensemble MAPE	Ensemble RMSE
RELIANCE.NS	1.18%	21.56
HDFCBANK.NS	1.16%	14.17
TATASTEEL.NS	1.68%	3.98
APOLLOHOSP.NS	1.08%	105.94
MARUTI.NS	1.45%	264.84

With ARIMA MAPE ≈ 0.9 – 1.2% vs. Prophet MAPE ≈ 4 – 27% , ARIMA captures ≈ 94 – 98% of effective ensemble weight. Ensemble MAPE is consequently above ARIMA alone — the small Prophet contamination degrades rather than improves accuracy. This is consistent with the Bates–Granger result that ensembles add value only when component forecast errors are complementary; here they are not.

5-Day Forecasts (Ensemble) Beyond 2026-05-12

Table 6: Ensemble day-1 and day-2 forecasts used for StockGro evaluation

Ticker	Day 1 (Rs.)	Day 2 (Rs.)
RELIANCE.NS	1,350.61	1,351.84
HDFCBANK.NS	770.90	772.09
TATASTEEL.NS	208.23	208.23
APOLLOHOSP.NS	7,936.45	7,926.45
MARUTI.NS	13,111.31	13,125.75

RMSE values are not comparable across tickers due to differing price scales.

Task 4 — Volatility and Trend Analysis

Volatility Estimation

$$\hat{\sigma}_{\text{ann}} = \text{std}(r_t) \cdot \sqrt{252} \quad \hat{\sigma}_{30d} = \text{std}_{30}(r_t)|_{t=T} \cdot \sqrt{252} \quad r_t = \ln(P_t/P_{t-1}) \quad (3)$$

Table 7: Volatility summary and STL trend classification

Ticker	Ann. Vol	Recent 30d	Change	Trend
RELIANCE.NS	27.36%	28.02%	+0.7 pp (exp.)	UP
HDFCBANK.NS	25.18%	28.82%	+3.6 pp (exp.)	UP
TATASTEEL.NS	35.15%	16.06%	−19.1 pp (comp.)	UP
APOLLOHOSP.NS	30.58%	17.33%	−13.2 pp (comp.)	UP
MARUTI.NS	28.03%	30.22%	+2.2 pp (exp.)	UP

The recent-volatility picture is mixed. TATASTEEL and APOLLOHOSP have undergone significant 30-day vol **compression** (−19 and −13 percentage points respectively), entering the forecast window in a calmer regime. By contrast HDFCBANK, RELIANCE, and MARUTI show mild **expansion**, with HDFCBANK’s 30-day vol (28.82%) now materially above its long-run average (25.18%) — likely reflecting heightened banking-sector uncertainty in early 2026. ARIMA confidence intervals, derived from full-period residual variance, will consequently be *conservative* (too wide) for the compression stocks and *slightly tight* for the expansion stocks — a known limitation of assuming homoscedastic residuals.

STL Trend Decomposition

STL with period = 252 and robust fitting decomposes: $P_t = T_t + S_t + R_t$. The normalised trend slope is computed via OLS on $\tilde{T}_t = T_t/T_0$:

$$\text{UP if } \hat{\beta}_1 > 5 \times 10^{-4}, \quad \text{DOWN if } \hat{\beta}_1 < -5 \times 10^{-4}, \quad \text{otherwise SIDEWAYS}$$

All five tickers satisfy the UP threshold, supporting a long-only allocation framework without defensive underweighting of any individual name.

Task 5 — Portfolio Construction and Capital Allocation

Strategy A — Forecast-Return Weights

$$\mu_i = \frac{\hat{P}_{i,T+5} - P_{i,T}}{P_{i,T}} \quad w_i^{\text{ret}} = \frac{\max(\mu_i, 0) + \varepsilon}{\sum_j [\max(\mu_j, 0) + \varepsilon]} \quad \varepsilon = 10^{-6} \quad (4)$$

HDFCBANK (+2.69%) is the sole stock with a positive 5-day forecast; it therefore absorbs essentially all of the return-weight pool (99.99%). RELIANCE (−0.87%), TATASTEEL (−1.78%), APOLLOHOSP (−1.03%), and MARUTI (−0.40%) carry negative expected returns and receive only the regularisation floor ε .

Strategy B — Inverse-Volatility Weights

$$w_i^{\text{vol}} = \frac{1/\hat{\sigma}_{i,30d}}{\sum_j 1/\hat{\sigma}_{j,30d}} \quad (5)$$

TATASTEEL’s dramatic 30d vol compression to 16.06% gives it the *highest* inverse-vol weight (27.87%); MARUTI’s mild expansion to 30.22% yields the lowest (14.81%).

Combined Final Allocation

$$w_i^* = \frac{0.5 w_i^{\text{ret}} + 0.5 w_i^{\text{vol}}}{\sum_j (0.5 w_j^{\text{ret}} + 0.5 w_j^{\text{vol}})} \quad (6)$$

Table 8: Final portfolio allocation — Rs. 10,00,000 capital

Ticker	Sector	Fcst Ret	Vol Wt	Ret Wt	Final Wt	Alloc (Rs.)
HDFCBANK.NS	Banking	+2.69%	15.53%	99.99%	57.76%	5,77,579
TATASTEEL.NS	Metals	−1.78%	27.87%	≈ 0	13.94%	1,39,370
APOLLOHOSP.NS	Healthcare	−1.03%	25.82%	≈ 0	12.91%	1,29,113
RELIANCE.NS	Energy	−0.87%	15.97%	≈ 0	7.99%	79,882
MARUTI.NS	Auto	−0.40%	14.81%	≈ 0	7.41%	74,056
Total					100.00%	10,00,000

HDFCBANK dominates at 57.76% as the only equity with a positive 5-day return forecast. TATASTEEL receives 13.94% driven purely by its inverse-volatility contribution — its 30-day vol compression to 16.06% makes it the lowest-risk name in the recent window. Portfolio-level metrics:

$$\mu_P = \sum_i w_i^* \mu_i = +1.07\% \quad \sigma_P = \sqrt{\mathbf{w}^{*\top} \mathbf{\Sigma} \mathbf{w}^*} = 21.68\% \text{ (ann.)} \quad (7)$$

The annualised portfolio volatility (21.68%) is materially below the equal-weighted average individual vol (≈ 29%), confirming diversification benefit from the five-sector composition despite the concentrated HDFCBANK position.

Task 6 — Model Comparison

Metrics

$$\text{MAPE} = \frac{1}{n} \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \quad (8)$$

$$\text{DirAcc} = \frac{1}{n} \sum \mathbf{1}[\text{sgn}(\hat{y}_t - y_{t-1}) = \text{sgn}(y_t - y_{t-1})] \times 100 \quad (9)$$

Full Comparison Table

Table 9: Walk-forward test-set performance — 226 observations per ticker

Ticker	ARIMA			Prophet		Ensemble	Naive
	MAPE	RMSE	DA	MAPE	RMSE	MAPE	MAPE
RELIANCE.NS	0.88	17.70	59.7%	14.52	235.09	1.18	0.89
HDFCBANK.NS	0.81	10.23	51.8%	12.60	147.54	1.16	0.81
TATASTEEL.NS	1.19	3.10	55.8%	26.60	54.96	1.68	1.19
APOLLOHOSP.NS	0.86	91.64	53.1%	4.17	370.52	1.08	0.86
MARUTI.NS	1.03	208.57	50.0%	18.24	3200.88	1.45	1.02

Discussion

ARIMA wins on MAPE and RMSE for all tickers. Its walk-forward discipline (226 refits on the test set) prevents in-sample optimism. Ljung-Box p -values ≥ 0.63 confirm no residual autocorrelation — the AIC-selected orders appear to capture most detectable linear structure.

Prophet performed substantially worse than ARIMA (MAPE 4–27%). Three root causes: (1) NSE equity prices in the Jul 2025 – May 2026 window do not exhibit the stable weekly/yearly seasonality Prophet

requires; (2) the piecewise-linear trend overshoots sharply after the structural downward move visible across Indian equities in H1 2026; (3) the conservative changepoint prior (0.05) prevents rapid adaptation to regime changes, amplifying lag errors. TATASTEEL’s 26.6% MAPE is particularly stark given that ARIMA achieves 1.19% on the same series with a (0, 1, 0) random walk — Prophet adds substantial noise.

Ensemble degrades accuracy vs. ARIMA alone. At $\approx 95\%$ ARIMA weighting, even a 5% Prophet contamination inflates MAPE by 0.22–0.49 pp. An ensemble is only beneficial when component errors are complementary; here both models tend to err in the same direction on trend moves, and Prophet’s much larger absolute errors dominate.

Naive baseline: ARIMA’s MAPE advantage over naive is 0.00–0.01 pp on MAPE, but 5–10 pp on directional accuracy (RELIANCE 59.7% vs 52.2%), which is the operationally relevant metric for trading decisions. The naive model cannot produce confidence intervals or directional probabilities.

Model chosen for portfolio: ARIMA was the primary signal. Ensemble forecasts were used for allocation (negligible practical difference given $>94\%$ ARIMA weighting in every case).

Task 7 — Virtual Trading on StockGro

Portfolio weights from Task 5 were translated into share quantities at Day 1 (14 May 2026) market prices.
Day 1: 14-05-2026, Day 2: 15-05-2026.

Table 10: StockGro execution log — Day 1 (14-05-2026) market close

Ticker	Wt (%)	Capital (Rs.)	Day-1 Price (Rs.)	Shares
HDFCBANK.NS	57.76	5,77,579	759.28	761
TATASTEEL.NS	13.94	1,39,370	217.10	642
APOLLOHOSP.NS	12.91	1,29,113	8,077.25	16
RELIANCE.NS	7.99	79,882	1,338.49	59
MARUTI.NS	7.41	74,056	12,901.16	5
Total	100.00	10,00,000	—	—

Transaction costs, slippage, and brokerage effects were ignored because StockGro operates as a virtual execution environment.

Task 8 — Performance Tracking and Comparison

Prediction vs. Actual

Day-1 and Day-2 actual closing prices were recorded from StockGro at 15:25 IST on each day. MAPE is computed per-stock per-day; directional accuracy compares the sign of the model’s forecast move to the sign of the actual move relative to the prior close.

Table 11: Forecast vs. actual closing prices and live prediction error

Ticker	Fcst D1 (Rs.)	Actual D1 (Rs.)	MAPE D1	Fcst D2 (Rs.)	Actual D2 (Rs.)
RELIANCE.NS	1,350.61	1,338.49	0.91%	1,351.84	1,321.81
HDFCBANK.NS	770.90	759.28	1.53%	772.09	762.63
TATASTEEL.NS	208.23	217.10	4.09%	208.23	214.52
APOLLOHOSP.NS	7,936.45	8,077.25	1.74%	7,926.45	7,971.38
MARUTI.NS	13,111.31	12,901.16	1.63%	13,125.75	12,988.44
Average	—	—	1.98%	—	—

An average Day-1 MAPE of **1.98%** on the live window is consistent with the walk-forward test-set MAPE range of 0.81–1.19% — the slightly higher live error is expected given the five forecasts are generated 5 days

ahead rather than 1 step ahead in the validation scheme. TATASTEEL (4.09%) is the worst performer, consistent with its random-walk (0, 1, 0) specification having near-zero predictive signal.

D1→D2 directional accuracy (model):

- RELIANCE: model predicts $\uparrow$ ($1351.84 > 1350.61$), actual $\downarrow$ ($1321.81 < 1338.49$) — **Wrong**
- HDFCBANK: model predicts $\uparrow$, actual $\uparrow$ — **Correct**
- TATASTEEL: model predicts flat ($\rightarrow\uparrow$), actual $\downarrow$ — **Wrong**
- APOLLOHOSP: model predicts $\downarrow$, actual $\downarrow$ — **Correct**
- MARUTI: model predicts $\uparrow$, actual $\uparrow$ — **Correct**

D1→D2 directional accuracy: $3/5 = 60\%$.

Portfolio Performance

Table 12: Two-day portfolio performance

Metric	Day 1 Close	Day 2 Close
HDFCBANK value (761 sh.)	Rs. 5,78,122	Rs. 5,80,562
TATASTEEL value (642 sh.)	Rs. 1,39,378	Rs. 1,37,782
APOLLOHOSP value (16 sh.)	Rs. 1,29,236	Rs. 1,27,542
RELIANCE value (59 sh.)	Rs. 78,971	Rs. 77,987
MARUTI value (5 sh.)	Rs. 64,506	Rs. 64,942
Portfolio Value (Rs.)	9,90,213	9,88,815
Absolute P&L (Rs.)	−9,787	−11,185
Portfolio Return (%)	−0.98%	−1.12%
Best contributor	HDFCBANK (+Rs. 2,440 D1→D2)	
Worst contributor	TATASTEEL (−Rs. 1,596 D1→D2)	

Day-1 value computed at actual D1 close prices; Day-2 at actual D2 close prices.

Reflection

Model alignment. The average live MAPE (1.98%) is marginally above the walk-forward test MAPE (0.81–1.19%) because the live forecasts are issued 5 days ahead rather than 1 step ahead. Despite this, ARIMA’s directional accuracy of 60% (3/5 stocks correct) on the live window exceeds the random baseline, consistent with its test-set performance.

HDFCBANK concentration. The strategy concentrated 57.76% of capital in HDFCBANK as the sole positive-return forecast. HDFCBANK rose slightly from Day 1 to Day 2 (+Rs. 3.35, +0.44%), partially offsetting losses in the other four positions. The model correctly predicted HDFCBANK’s upward direction — validating the concentration decision even though the overall portfolio was slightly negative.

TATASTEEL anomaly. TATASTEEL’s (0, 1, 0) forecast predicted flat prices (Rs. 208.23 for all 5 days) while the actual price moved upward through Rs. 217. A random-walk model has near-zero directional predictive power, making this stock less suitable for directional forecasting under the current specification. Its 13.94% allocation was driven entirely by its low recent volatility, not by any forecast confidence.

Vol estimation accuracy. HDFCBANK’s 30-day vol was rising (+3.6 pp expansion noted in Task 4), yet its actual D1→D2 move (+0.44%) was modest — suggesting the expansion was driven by a brief spike that had already partially resolved. ARIMA’s full-period confidence intervals were therefore appropriately conservative.

What would change in future iterations. (i) Replace Prophet with Holt–Winters exponential smoothing for a more suitable seasonal competitor on financial data. (ii) Add GARCH(1,1) for forward-looking volatility to make inverse-vol weights predictive rather than backward-looking — critical given the large spread between compression (TATASTEEL) and expansion (HDFCBANK) observed in Task 4. (iii) Incorporate the pre-computed correlation matrix (`correlation_matrix.csv`) into a Markowitz mean–variance step to explicitly

penalise co-movement risk. (iv) Use BIC instead of AIC in the order grid search to prefer parsimonious models, especially given TATASTEEL degenerating to $(0, 1, 0)$ already suggests over-parameterisation is not the main risk.

Appendix — Residual Diagnostics

Table 13: ARIMA residual diagnostics — final model on full dataset (train + test)

Ticker	Order	JB p	LB p (lag 10)	Interpretation
RELIANCE.NS	(2, 1, 2)	≈ 0	0.9998	No autocorr; fat-tailed residuals
HDFCBANK.NS	(2, 1, 2)	≈ 0	0.9938	No autocorr; fat-tailed residuals
TATASTEEL.NS	(0, 1, 0)	≈ 0	0.9335	No autocorr; fat-tailed residuals
APOLLOHOSP.NS	(2, 1, 2)	≈ 0	0.6364	No autocorr; fat-tailed residuals
MARUTI.NS	(2, 1, 2)	≈ 0	0.9998	No autocorr; fat-tailed residuals

Ljung-Box $p \gg 0.05$ for all tickers confirms failure to reject H_0 (no serial autocorrelation): the AIC-selected orders successfully extracted most detectable linear structure. JB $p \approx 0$ universally confirms fat-tailed, non-Gaussian residuals — a well-known property of equity returns. This means Gaussian 95% confidence intervals are technically narrower than empirically warranted but does not impair point-forecast accuracy. Full residual plots (histogram + ACF) are saved to `outputs/residual_diagnostics.png`.